

# Circles

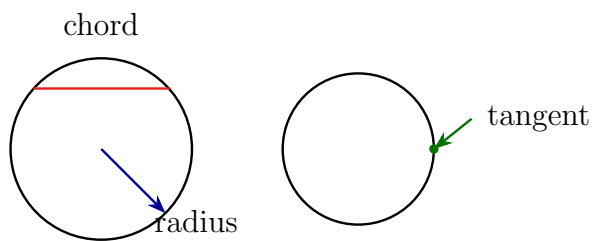
US Common Core – High School Geometry

Domain: Circles (G-C.A) | Expressing Geometric Properties with Equations (G-GPE.A)

## 1 Basic Vocabulary

### Definitions

- A **circle** is a closed figure with every point the same distance from one fixed point, the **center**.
- The **radius** is the distance from the center to any point on the circle; the **diameter** is twice the radius.
- A **chord** is a segment joining two points on the circle.
- A **tangent** line touches the circle at exactly one point, the **point of tangency** (point of contact).
- A **major segment** and **minor segment** are the two regions a chord divides a circle into (major = larger, minor = smaller).
- A **sector** is the pie-slice region bounded by two radii and an arc; the angle between the radii is the **central angle**.



## 2 Two Circles Together

### Definitions

- **Intersecting circles:** two circles crossing at two points.
- **Tangent circles:** two circles meeting at exactly one point.
- **Concentric circles:** two circles with the *same* center but different radii.

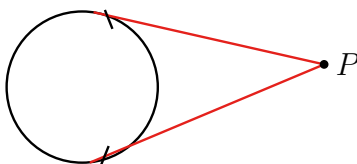
### 3 Tangent Line Theorems

#### Theorem 1

Any tangent to a circle is always perpendicular to the radius drawn to the point of tangency.

#### Theorem 2 (Two Tangent Segments Theorem)

Tangent segments to a circle from the same external point are always congruent to each other.



#### Example

Two tangent segments from an external point measure  $(5x + 2)$  and  $(78 - x)$ :

$$5x + 2 = 78 - x \Rightarrow 6x = 76 \Rightarrow x = \frac{38}{3}$$

### 4 Arcs

#### Sector and Sectoral Angle

A **minor arc/sector** is the smaller of the two regions; the **major arc/sector** is the larger. The angle at the center between the two bounding radii is the **sectoral (central) angle**.

#### Theorem 1 (Arc Addition Postulate)

If  $B$  lies between  $A$  and  $C$  on a circle, then

$$m\widehat{ABC} = m\widehat{AB} + m\widehat{BC}.$$

#### Theorem 2 (Congruent Circles)

Any two circles are congruent with each other if and only if both circles have the same radius.

#### Similar Circles Theorem

All circles are similar to each other.

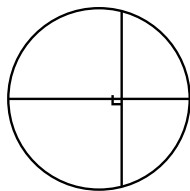
## 5 Chords

### Perpendicular Bisector Theorem (Chords)

If the diameter of a circle is perpendicular to a chord, then the diameter bisects the chord.

### Converse

If the diameter of a circle is bisecting a chord, then the diameter is perpendicular to the chord.



### Corresponding Minor Arc Theorem

In the same circle, or in congruent circles, the minor arcs are congruent with each other if and only if the chords are congruent with each other.

### Congruent Chord Theorem

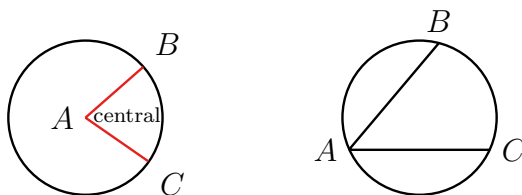
In the same circle, two chords are congruent with each other if and only if both chords are equidistant from the center.

## 6 Inscribed Angles and Polygons

### Theorem 1 (Inscribed Angle Theorem)

The measure of an inscribed angle is half the measure of its intercepted arc:

$$m\angle A = \frac{1}{2} m\widehat{BC}$$



### Theorem 2

Inscribed angles that intercept the same arc are congruent to each other.

**Theorem 3**

An inscribed angle subtending a semicircle (i.e. standing on a diameter) is always a right angle.

**Theorem 4 (Inscribed Quadrilateral Theorem)**

A quadrilateral can be inscribed in a circle if and only if its opposite angles are supplementary.

## 7 Angle Relationships in a Circle

### Tangent-Chord Angle Theorem

The angle between a tangent and a chord drawn from the point of tangency is half the intercepted arc:

$$m\angle 1 = \frac{1}{2} m\widehat{AB}, \quad m\angle 2 = \frac{1}{2} m\widehat{ABC}$$

### Where Two Lines Can Meet a Circle

Two lines through a circle can intersect **on** the circle, **inside** the circle, or **outside** the circle – each case has its own angle rule below.

### Angles Inside the Circle (Two Chords)

If two chords intersect inside a circle, the measure of the angle formed is half the sum of the two intercepted arcs:

$$m\angle 1 = \frac{1}{2} (m\widehat{DC} + m\widehat{AB}), \quad m\angle 2 = \frac{1}{2} (m\widehat{BC} + m\widehat{AD})$$

### Angles Outside the Circle (Secants / Tangents)

If two secants, two tangents, or a secant and a tangent meet at a point outside the circle, the angle formed is half the *difference* of the two intercepted arcs:

$$m\angle = \frac{1}{2} (\text{far arc} - \text{near arc})$$

### Circumscribed Angle Theorem

If  $D$  is an external point with two tangents touching the circle at  $A$  and  $B$  (center  $C$ ), then

$$m\angle ADB = 180^\circ - m\angle ACB.$$

## 8 Segment Relationships in a Circle

### Segment Names (from an external point)

A **tangent segment** touches the circle at one point; a **secant segment** crosses through the circle; the **external secant segment** is the part of a secant outside the circle, from the external point to the near intersection.

### Theorem 1 (Two Chords)

If two chords intersect inside a circle at  $E$ :

$$EA \cdot EB = EC \cdot ED$$

### Theorem 2 (Tangent-Secant)

If a tangent and a secant are drawn from the same external point, the tangent length squared equals the product of the whole secant and its external part:

$$(\text{tangent})^2 = (\text{near secant segment}) \times (\text{whole secant segment})$$

*Note: The source notes write this case as  $AB \cdot BC = AD^2$ ; based on the figure ( $AD$  tangent,  $AB$  the external part and  $AC$  the whole secant), the standard and verifiable form of this theorem is  $AD^2 = AB \cdot AC$ , shown above in words to avoid ambiguity in relabeling.*

### Theorem 3 (Two Secants)

If two secants are drawn from the same external point  $A$ , each crossing the circle at a near and far point:

$$AB \cdot AC = AD \cdot AE$$

## 9 The Equation of a Circle

### Standard Form

A circle with center  $C = (h, k)$  and radius  $r$  has equation

$$(x - h)^2 + (y - k)^2 = r^2.$$

